Advanced General Relativity

Master AFPC @ UB

ICCUB
Institut de Ciències del Cosmos
UNIVERSITAT DE BARCELONA
EXCELENCIA MARÍA DE MAEZTU 04/2025-03/2031
MINISTERIO DE CIENCIA, INNOVACIÓN Y UNIVERSIDADES
AGENCIA ESTATAL DE INVESTIGACIÓN

A course on the fundamentals of

Gravitation and Spacetime

1. **Black Holes – Classical and Quantum**
2. **Inflation and the Origin of Cosmic Structure**
3. **Gravitational Waves – A New Window for Exploration**

Roberto Emparan

General Formalism
Black Holes
Quantum Gravity

6 weeks



Alessio Notari

Relativistic Cosmology
Cosmological perturbations
Inflation

4 weeks



Ruxandra Bondarescu

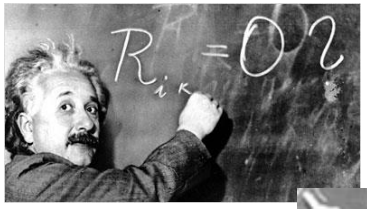
Gravitational wave astronomy
GW from black holes and neutron stars
From the stellar graveyard to the universe at large

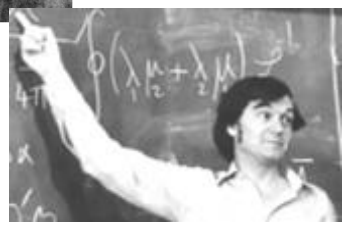
2 weeks



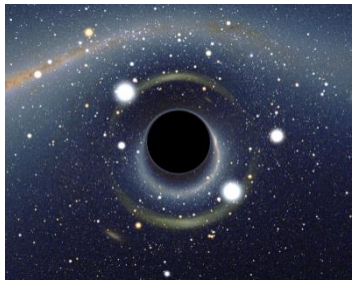
Pre-requisite: Introductory course to General Relativity

What will you learn?

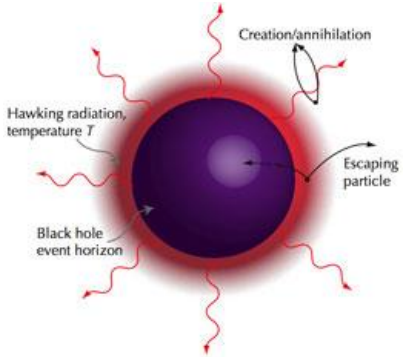




- Advanced techniques in GenRel
  - Lagrangian formulation, Causal Structure (Horizons)...

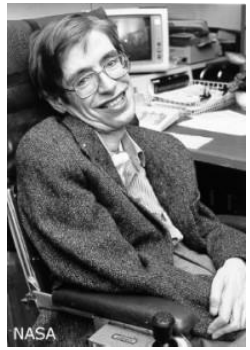


- Black Holes
  - Classical theory
  - Quantum effects – Hawking radiation
  - Black hole thermodynamics

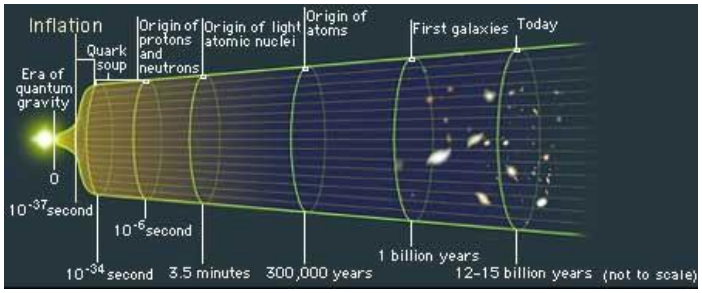


- Basic notions about

  Quantum theory of Gravity

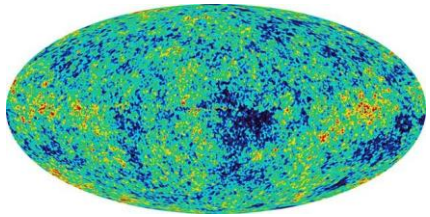


What will you learn?



- Relativistic Cosmology
- Cosmological perturbation theory

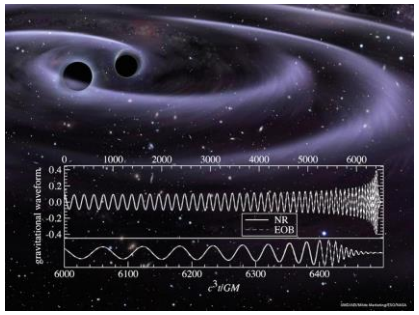
  -Cosmic Microwave Background (CMB) & Large Scale Structure
- Inflation as a theory of initial conditions
  - Quantum origin of primordial perturbations

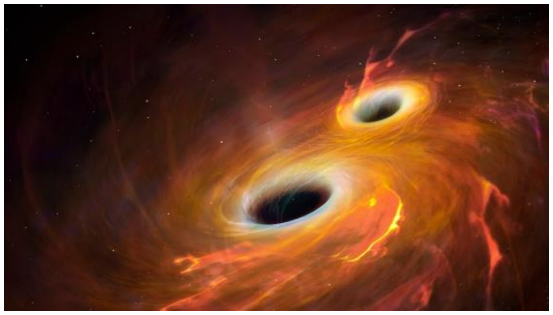


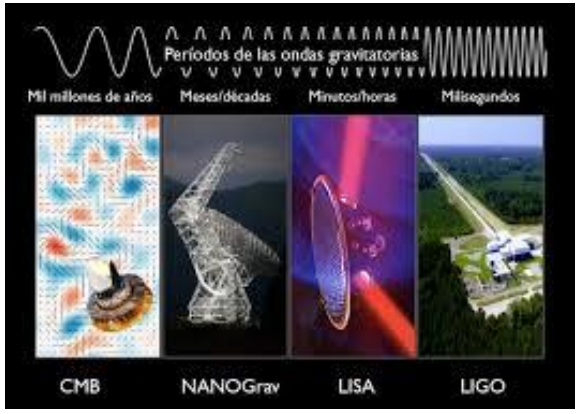
What will you learn?

- What are the known sources of gravitational waves
- How to compute the gravitational wave emitted by

  (a) a binary inspiral, (b) a perturbed neutron star
- Mining data and catching gravitational waves
- How to draw conclusions about the universe at large from the LIGO-Virgo-KAGRA catalogue







What is the course like?

- Theoretical – be ready for math
- On blackboard – we'll go into details
- Homework (weekly)

  complement to lectures

  essential practice

  evaluation*

* alternatively, exam if requested